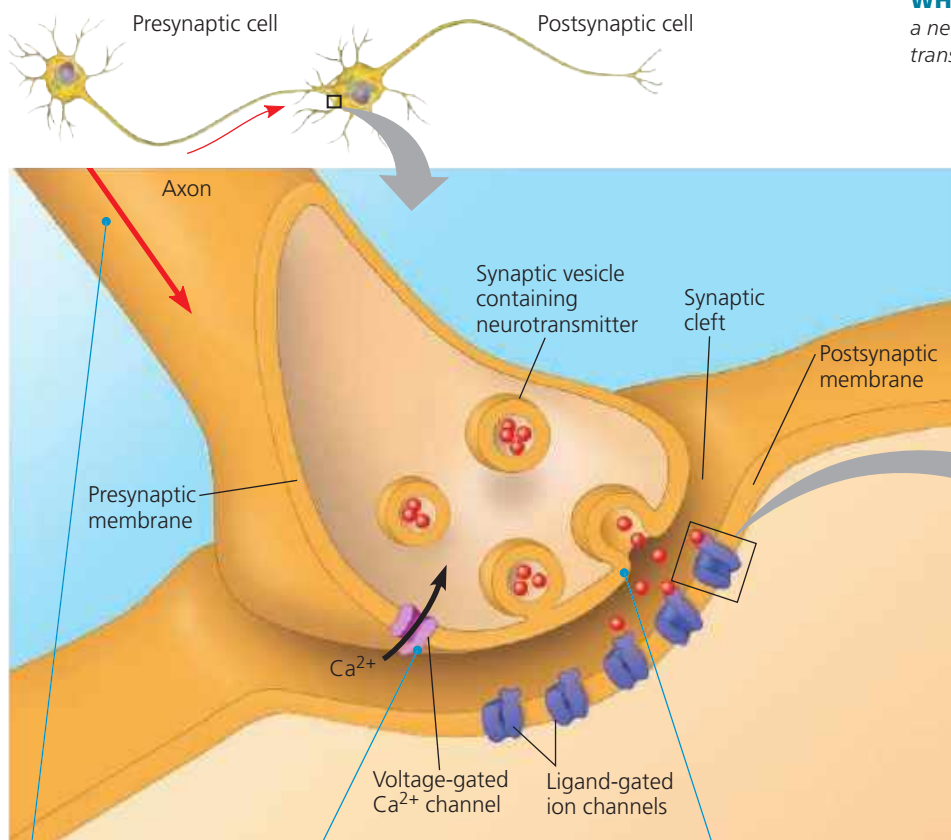
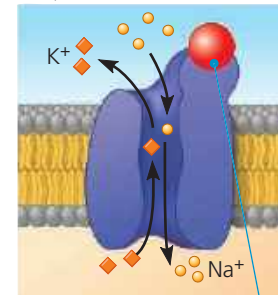


▼ **Figure 48.16 A chemical synapse.** This figure illustrates the sequence of events that transmits a signal across a chemical synapse. In response to binding of neurotransmitter, ligand-gated ion channels in the postsynaptic membrane open (as shown here) or, less commonly, close. Synaptic transmission ends when the neurotransmitter diffuses out of the synaptic cleft, is taken up by the synaptic terminal or by another cell, or is degraded by an enzyme.



WHAT IF? If all the Ca^{2+} in the fluid surrounding a neuron were removed, how would this affect the transmission of information within and between neurons?

➔ **Mastering Biology BioFlix® Animation: How Synapses Work**
HHMI Videos: Baldomero Olivera on the Deadly *Conus geographicus* • Blockage of Motor Neuron Synapses by Cone Snail Toxins



1 An action potential arrives, depolarizing the presynaptic membrane.

2 The depolarization opens voltage-gated channels, triggering an influx of Ca^{2+} .

3 The elevated Ca^{2+} concentration causes synaptic vesicles to fuse with the presynaptic membrane, releasing neurotransmitter into the synaptic cleft.

4 The neurotransmitter binds to ligand-gated ion channels in the postsynaptic membrane. In this example, binding triggers opening, allowing Na^+ and K^+ to diffuse through.

Information transfer at chemical synapses can be modified by altering either the amount of neurotransmitter that is released or the responsiveness of the postsynaptic cell. Such modifications underlie an animal's ability to alter its behavior in response to change and also form the basis for learning and memory, as you will read in Concept 49.4.

Generation of Postsynaptic Potentials

At many chemical synapses, the receptor protein that binds and responds to neurotransmitters is a **ligand-gated ion channel**, often called an *ionotropic receptor*. These receptors are clustered in the membrane of the postsynaptic cell, directly opposite the synaptic terminal. Binding of the neurotransmitter (the receptor's ligand) to a particular part of the receptor opens the channel and allows specific

ions to diffuse across the postsynaptic membrane. The result is a *postsynaptic potential*, a graded potential in the postsynaptic cell.

At some chemical synapses, the ligand-gated ion channels are permeable to both K^+ and Na^+ (see Figure 48.16). When these channels open, the membrane potential depolarizes toward a value roughly midway between E_{K} and E_{Na} . Because such a depolarization brings the membrane potential toward threshold, it is called an **excitatory postsynaptic potential (EPSP)**.

At other chemical synapses, the ligand-gated ion channels are selectively permeable for only K^+ or Cl^- . When such channels open, the postsynaptic membrane hyperpolarizes. A hyperpolarization produced in this manner is an **inhibitory postsynaptic potential (IPSP)** because it moves the membrane potential farther from threshold.